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Transformers: Applications in Language and Communication

Final Report

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The Effects of Emotional Framing of Factual Questions on Hallucination Rates for Multiple Transformer-Based LLMs

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1 Introduction

The advancement of natural language processing (NLP) and natural language generation (NLG) has led to remarkable achievements in the deployment of transformer-based large language models (LLMs), such as GPT-4 or Claude. This has shaped how users access and process information in various domains for different tasks. Although these models are very powerful demonstrating impressive performance across tasks like question answering, summarization and dialogue generation, they do suffer from a persistent limitation: the tendency to produce "hallucination". These hallucinations are defined as plausible yet factually incorrect or fabricated information (Alansari & Luqman, 2024). Such phenomena can pose high risks in domains that require factual accuracy such as education, healthcare and information retrieval, users simply accepting and trusting these generated hallucinations as factually correct information Kaate et al. (2025). As a result, ensuring the reliability of LLM generated content has become a central concern in both academic research and real-world applications.

Existing research has made significant progress in identifying the causes for hallucinations and proposed strategies to mitigate them. Alansari and Luqman (2024) argue that hallucination can be present at any stages of the LLM development pipeline from training data, model architecture to inference methods. They further argue that once deployed, one main cause of hallucination is ambiguous prompting. This is further supported by Li et al. (2023) who reveal that LLMs exhibit unexpected sensitivity to prompt framing, with performance varying substantially based on what may seem just superficial textual modifications. They showed that incorporating emotional phrases into prompts can improve LLM performance on deterministic tasks. This finding suggests that emotional framing may modulate model behaviour, however, the relationship between emotional prompting and hallucination remains largely unexplored. While emotional framing can enhance performance on some tasks, it remains unclear whether it also influences the accuracy of factual recall or if it increases tendencies towards fabrication when models are prompted with uncertain queries.

Within this context, this study wishes to understand the role of emotional framing in factual queries. In real-world interactions, users frequently embed questions with emotionally charged language, expressing urgency, concern or personal relevance. We hypothesize that prompting LLMs models with emotional cues may lead to higher rates of hallucinations, particularly when combined with models' tendency to generate confident responses even when uncertain (Xiong et al., 2023). More specifically, this study addresses the following research question:

RQ: Does the emotional framing of a factual question influence hallucination rates in transformer-based LLMs?

Investigating this question is both practically and theoretically relevant. Theoretically, it contributes to better understanding how contextual and affective cues interact with probabilistic language processes. Practically, it helps in the advancement of prompt engineering, model evaluation and safe deployment of LLMs. As LLMs increasingly persevere in our everyday communication, understanding how emotionally framed queries impact their reliability is essential in mitigating risks and improving the trustworthiness of such models. In order to tackle this research question, we created a dataset of 823 validated stimuli, each representing factual questions from the TriviaQA dataset (Joshi et al., 2017) which were further enriched with emotional framings. This dataset was then used to evaluate the hallucination rate of five different transformer-based models: Flan-T5-large, Mistral-7B-Instruct-v0.3, Falcon-7b-instruct, Meta-Llama-3-8B-Instruct and Phi-3-mini-4k-instruct.

2 Literature Review

2.1 Hallucinations in Large Language Models

Hallucination in LLMs is a generated output which may seem correct and coherent with the prompted questions by users, but which is in fact inaccurate or false. More specifically, Huang et al. (2023) distinguish between two types of hallucinations: Factual hallucination – LLMs outputs can either be factually correct but contradicting or pure fabrications due to overstatements or simply unverifiable information. Faithfulness hallucination – LLMs deviate from the user prompted instructions by completely neglecting the user input, by changing the context or by logical fallacy. Recent theoretical work shows that hallucinations are an inherent limitation rather than just an engineering challenge. Xu et al. (2024) demonstrate that hallucinations are inevitable in LLMs as, mathematically, no finite model is able to learn all computable functions. This suggests that the complete removal of hallucinations may be fundamentally impossible and therefore only the detection and the use of mitigation strategies can help improve the trustworthiness of LLM models.

In order to identify hallucinations, a variety of detection methods have been explored from simple consistency checks to more sophisticated approaches. Farquhar et al. (2024) propose the use of semantic entropy in detecting confabulations – a form of hallucinations – without requiring external knowledge bases. Their method achieved significant improvements in question-answering tasks and in generation of longer texts such as biographies. Their findings suggest that models possess implicit representation of their own knowledge limitations even when they generate confident but incorrect responses.

When it comes to mitigation strategies to aid LLMs in generating more correct outputs, Dhuliawala et al. (2023) proposed the chain-of-verification (CoVe) prompting method. This prompting technique requires the models to first generate an initial response that may contain hallucinations, then systematically verify factual claims through independent question-answering

before producing the final outputs. This self-correcting approach is shown to reduce hallucination rates across multiple benchmarks without requiring external retrieval systems. However, Gekhman et al. (2024) present a paradoxical concern: fine-tuning LLM models on new factual knowledge can increase hallucination rates over time, often ending up generating confident fabrications instead of properly incorporating new knowledge.

2.2 Emotional Prompting and Sycophantic Behaviour

Research has also been conducted in evaluating how emotional prompting can influence AI systems in recognizing, interpreting and responding. Li et al. (2023) propose a framework (EmotionPrompt) which incorporates psychologically grounded emotional stimuli into prompts. They argue that including established psychological phenomena based on cognitive theories into prompts can increase LLMs performance. They demonstrate that LLMs performance improves between 8-115% when prompts are more emotionally driven across tasks like instruction following, mathematical reasoning or social reasoning. Specifically, the larger a LLM is, the more emotional prompting tends to increase model performance. These findings enforce that emotional framing influences how LLMs behave by implicitly motivating the system to perform better ("Are you sure that's your final answer?") rather than explicit instructions. Moreover, research in affective computing demonstrates that LLMs are increasingly designed to detect and generate emotionally conditioned language, treating emotional signals as meaningful contextual inputs rather than superficial stylistic features (Zhang et al., 2024). These systems are often optimized to align with user intent and emotional tone, which may come to the expense of strict factual accuracy. Sharma et al. (2023) argue that such phenomena, known as sycophancy – the tendency of LLMs to conform to user opinions even when those opinions are incorrect – leads to biased models which tend to prioritize helpfulness and agreeableness over truthfulness in their responses.

Building on this line of research, Gozzi and Fallucchi (2026) show that emotional framing affects objective task performance for different LLMs.

By varying emotional tone while keeping semantic content constant, they found that positive and neutral prompts yield higher accuracy, while more negatively framed prompts lead to lower performance. These consistent effects suggest that emotional cues influence how LLMs process and generate responses. However, a clear relation between hallucination rates and emotional framing of factual information remains still largely unexplored, despite the risk of these architectures to conform with the user's inputs over factual correctness.

This study aims to build upon previous streams of research by examining whether emotional framing – shown to enhance performance on some tasks – also influences factual accuracy and hallucination rates. By controlling for domain effects and using validated emotional stimuli across a standardized factual QA benchmark, we provide empirical evidence addressing whether affective prompt engineering represents a viable strategy for improving LLM reliability or whether it introduces additional risks through sycophancy-like conformity mechanisms.

3 Methodology

3.1 Research Design

This study investigates whether the emotional framing of a factual question influences hallucination rates in transformer-based large language models (LLMs). Five publicly available instruction-tuned models were evaluated:

- `google/flan-t5-large` (780M parameters, encoder-decoder architecture),
- `mistralai/Mistral-7B-Instruct-v0.3` (7B),
- `tiiuae/falcon-7b-instruct` (7B, fp16),
- `meta-llama/Meta-Llama-3-8B-Instruct` (8B), and
- `microsoft/Phi-3-mini-4k-instruct` (3.8B, GGUF q4).

Models were selected to span a range of architectures, parameter scales, and training regimes, allowing cross-model comparison while maintaining a consistent evaluation protocol.

3.2 Dataset Construction

To investigate the effect of emotional framing on hallucination rates, we constructed a new dataset of emotionally reframed factual questions, as no existing dataset combines factual ground truth with controlled emotional framing.

3.3 Source Data

240 factual questions were sampled from TriviaQA (Joshi et al., 2017) using the `rc.wikipedia` configuration, which provides Wikipedia-verified answers with multiple accepted aliases. Questions were stratified across four topic domains: geography, history, science, and culture with 60 questions per domain, and filtered to a maximum of 20 words to ensure self-contained questions suitable for emotional rewriting.

3.4 Emotional rewriting

Each source question was rewritten into four emotional registers: joy, sadness, anger, and fear. This produced 960 rewritten variants alongside 240 neutral (unmodified) questions, for a total of 1,200 stimuli. Rewriting was performed using the AnthropicAPI with the prompt shown in the **Appendix A**.

3.5 Stimulus validation

Each rewritten question was validated using a pretrained GoEmotions classifier (`SamLowe/roberta-base-go-emotions`) to verify that the intended emotional signal was detectable in the stimulus. GoEmotions' 28 fine-grained labels were mapped to five coarse conditions, and a stimulus was accepted if the summed probability mass for the target emotion exceeded a threshold of 0.30. Stimuli that failed validation were flagged and excluded from the experimental analysis.

After validation-based filtering, the final dataset comprises 823 question-answer pairs. The emotional condition distribution reflects attrition from the validation step, with neutral (n = 240) and sadness (n = 200) being the most represented conditions and joy the least (n = 80). Topic domain coverage remained balanced across geography (n = 210), history (n = 203), science (n = 204), and culture (n = 206). Each record includes the original question, the emotionally rewritten variant, the ground-truth answer, a list of valid aliases, the topic domain, and a per-stimulus validation score. The mean validation score across accepted stimuli was ≥ 0.95, confirming that retained rewrites reliably conveyed their intended emotional register.

3.6 Inference Protocol

Each model received every stimulus in the dataset exactly once via a shared inference function, `run-inference-with-checkpointing`, which iterates over the dataframe, invokes a model-specific generation function, and saves progress to a CSV checkpoint file every 50 rows. This checkpointing strategy al-

lowed recovery from GPU memory interruptions without re-running completed inference passes. Generation was performed in greedy decoding mode (`do-sample=False`) with a maximum of 50 new tokens per response, constraining outputs to short factual answers and reducing free-form verbosity.

The prompt format differed by architecture. For FLAN-T5, which does not support chat templates, a direct instruction was prepended to the question: the model received a single string instructing it to answer in as few words as possible with no preamble. For the four causal decoder models (Mistral, Falcon, Llama-3, Phi-3), each model's native chat template was applied via the tokenizer's `apply-chat-template` method with a system message establishing the factual QA role, ensuring that instruction formatting followed each model's training conventions.

3.7 Response Scoring

Model outputs were scored through a four-layer correctness pipeline implemented in the shared `score-response` function. Layers were applied in sequence, and a response was marked correct as soon as any layer matched.

Layer 1 Exact match. The model response was compared case-insensitively against the ground-truth answer and all accepted aliases.

Layer 2 Alias containment. A whole-word regular expression search checked whether any accepted answer appeared as a substring within the model response, handling cases where models produced correct answers embedded in longer output strings.

Layer 3 Normalised match. Both the response and each candidate answer were lowercased, stripped of articles (a, an, the) and punctuation, and collapsed to single whitespace before comparison. This layer caught surface-form variation such as inconsistent capitalisation or trailing punctuation.

Layer 4 Semantic similarity. Responses that passed none of the lexical layers were encoded alongside all candidate answers using `all-MiniLM-L6-v2`,

a lightweight sentence transformer. Cosine similarity was computed between the response embedding and each candidate embedding. A response was considered correct if the maximum similarity score met or exceeded a threshold of 0.75. This final layer captured paraphrastic correct answers that share no surface-level token overlap with the reference.

Responses classified as incorrect were further categorised by the *classify-hallucination* function into three error types: substitution (a short, confident but wrong answer, $\leq$ 15 words), confabulation (a longer, elaborated incorrect response, > 15 words), and uncertainty (responses containing hedging language such as "I don't know" or "I'm not sure"), with a fourth category, refusal or empty, reserved for blank or near-blank outputs.

3.8 Statistical and Qualitative Analysis

Hallucination rates were computed per model as the proportion of incorrect responses across all 823 stimuli. Condition-level rates were then disaggregated by emotion condition and topic domain. The deviation from the neutral baseline, referred to throughout as delta from neutral, was computed by subtracting each condition's hallucination rate from the neutral hallucination rate for that model.

To test for a statistically significant association between emotional framing and hallucination outcome, Pearson's chi-squared test of independence was applied to a 5 × 2 contingency table (five emotion conditions × correct/incorrect) for each model. Effect size was quantified using Cramér's V, interpreted under thresholds: V < 0.10 negligible, 0.10–0.20 small, 0.20–0.40 medium, $\geq$ 0.40 large. An identical procedure was applied to a 4 × 2 domain-level contingency table. Pairwise domain comparisons were conducted using chi-squared tests on 2 × 2 subtables, with Bonferroni correction applied across all six domain pairs (corrected $\alpha = 0.05 / 6 = 0.0083$). All statistical tests were implemented using `scipy.stats.chi2-contingency`.

In addition to aggregate statistics, a qualitative analysis identified cases where a model answered correctly under the neutral condition but produced an incorrect response under one or more emotional conditions for

the same underlying question. These cases represent direct, within-question evidence of emotion-induced degradation, where emotional framing is the likely source of the error. For each such case, the error type (substitution or confabulation) and domain were recorded, and representative examples were selected for examples.

4 Results

4.1 Overall Performance

Model performance varied substantially across the five evaluated models (Table 1). Mistral-7B and Llama-3-8B-Instruct performed nearly identically, achieving the highest accuracies at 74.5% and 74.2% respectively, both with neutral baseline hallucination rates of 25.0%. Phi-3 Mini occupied the middle range at 60.8% accuracy, while Falcon-7B-Instruct (40.7%) and Flan-T5-Large (21.4%) showed considerably weaker factual reliability. Notably, Flan-T5-Large hallucinated on nearly four out of five questions, suggesting fundamental limitations in its factual knowledge base. Figure 1 illustrates this performance spread.

Metric	**Mistral-7B**	**Llama-3-8B**	**Phi-3 Mini**	**Falcon-7B**	**Flan-T5-Large**
Accuracy (%)	74.5	74.2	60.8	40.7	21.4
Hallucination Rate (%)	25.5	25.8	39.2	59.3	78.6
Neutral Baseline HR (%)	25.0	25.0	37.5	57.1	78.8

Table 1: Overall accuracy, hallucination rate, and neutral baseline hallucination rate for each model.

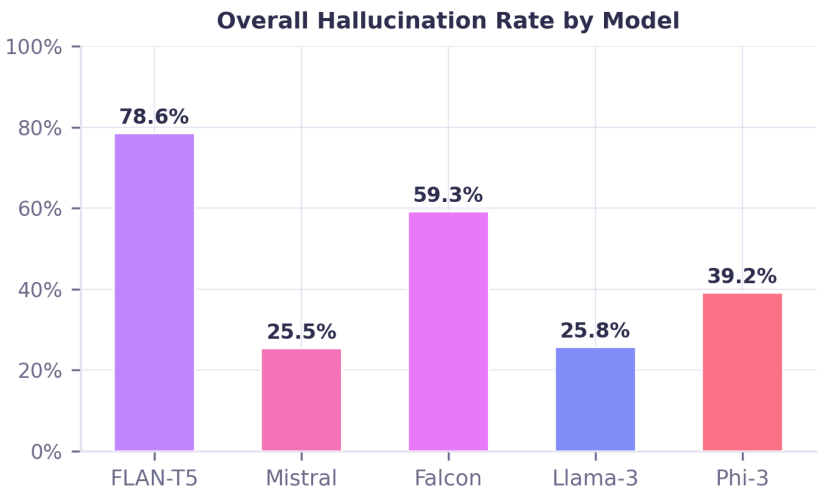

Figure 1: Overall hallucination rate by model.

4.2 Effect of Emotional Framing

Across all five models, emotional framing produced only modest shifts in hallucination rates relative to neutral baselines (Table 2, Figure 2). No model exhibited statistically significant differences across emotion conditions. However, a consistent directional pattern emerged: anger was the only emotion to reduce hallucination rates below neutral in four out of five models, with Mistral, Llama-3, and Phi-3 all showing decreases of approximately 3.6–3.8 percentage points under anger framing (Figure 3). Conversely, sadness and fear tended to increase hallucination rates, with the largest deviation observed in Falcon under joy framing (+7.9 pp). Flan-T5-Large was the exception, showing near-uniform hallucination rates across all emotional conditions, suggesting insensitivity to emotional cues entirely. As shown in Figure 5, the Cramér's V effect sizes for emotion were negligible across all models (ranging from 0.020 to 0.058), confirming the limited practical significance of emotional framing.

Emotion	**Mistral-7B**	**Llama-3-8B**	**Phi-3 Mini**	**Falcon-7B**	**Flan-T5-Large**
Neutral	25.0	25.0	37.5	57.1	78.8
Joy	27.5 (+2.5)	26.2 (+1.2)	41.2 (+3.7)	65.0 (+7.9)	77.5 (−1.2)
Sadness	26.5 (+1.5)	27.0 (+2.0)	42.0 (+4.5)	62.0 (+4.9)	79.5 (+0.8)
Anger	21.2 (−3.8)	21.2 (−3.8)	33.9 (−3.6)	56.8 (−0.3)	77.1 (−1.6)
Fear	27.0 (+2.0)	28.1 (+3.1)	41.1 (+3.6)	58.4 (+1.3)	78.9 (+0.2)

Table 2: Hallucination rate (%) by emotional condition for each model. Deviations from the neutral baseline are shown in parentheses.

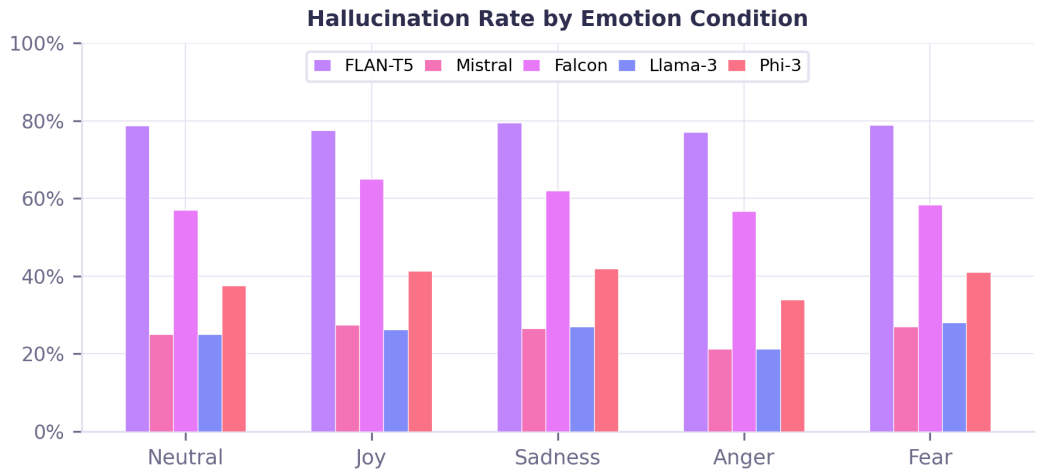

Figure 2: Hallucination rate by emotion condition across all models.

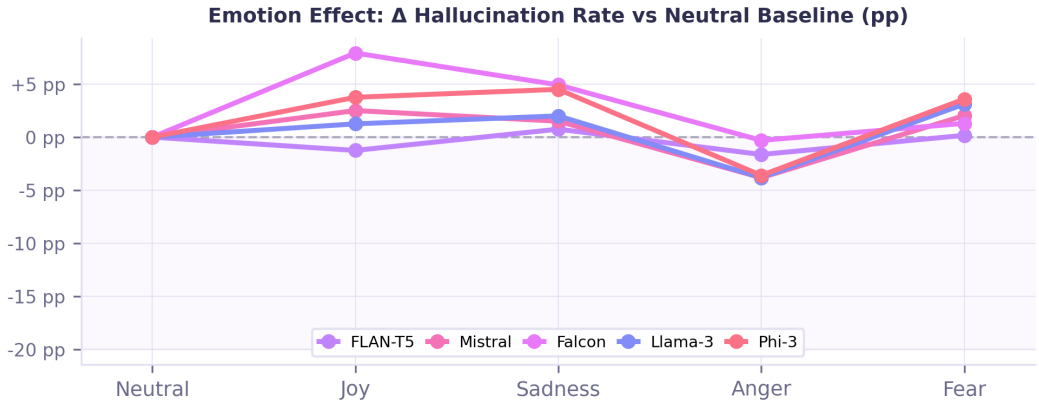

Figure 3: Change in hallucination rate (percentage points) relative to the neutral baseline for each emotion condition.

4.3 Effect of Topic Domain

In contrast to the modest emotion effects, topic domain had a strong and statistically significant influence on hallucination rates across all models (Table 3, Figure 4). Geography was consistently the best-performing domain, with hallucination rates ranging from 13.8% (Mistral and Llama-3) to 63.3% (Flan-T5). Culture and history were the most error-prone domains across all models. The most extreme case was Flan-T5's performance on culture questions, where 99.0% of responses were hallucinated, indicating a near-complete gap in domain knowledge rather than a framing effect. The domain effect was also substantially larger in magnitude than the emotion effect for every model tested, with cross-domain spreads of 22–36 percentage

points compared to cross-emotion spreads of only 2–8 percentage points. This contrast is clearly reflected in the Cramér's V effect sizes (Figure 5), where domain effects (0.186–0.316) consistently exceeded emotion effects (0.020–0.058) by a factor of approximately 4 to 16.

Domain	Mistral-7B	Llama-3-8B	Phi-3 Mini	Falcon-7B	Flan-T5-Large
Geography	13.8	13.8	28.6	40.5	63.3
Science	28.9	20.1	31.4	63.7	74.5
Culture	23.8	30.6	50.0	68.0	99.0
History	36.0	38.9	47.3	65.5	77.8

Table 3: Hallucination rate (%) by topic domain for each model.

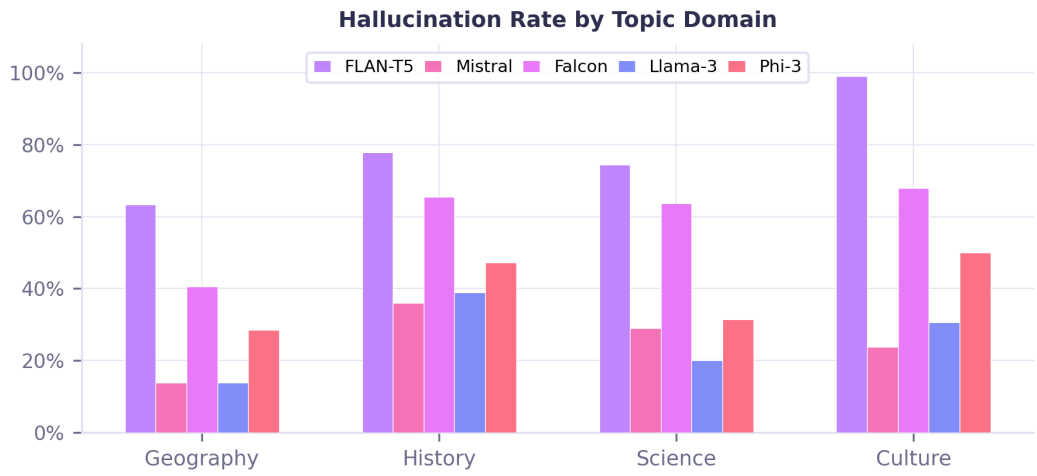

Figure 4: Hallucination rate by topic domain across all models.

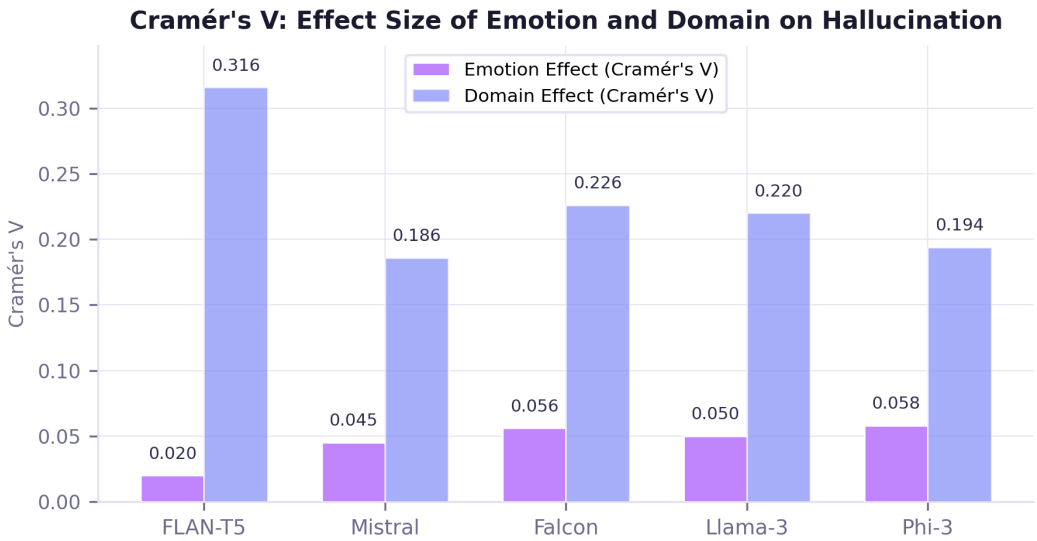

Figure 5: Cramér's V effect sizes for emotion and domain effects on hallucination across all models.

4.4 Emotion × Domain Interaction

The interaction between emotional framing and topic domain reveals that emotion effects are not uniform across knowledge areas (Figure 6). In domains where models already exhibit high hallucination rates, such as history and culture, the additional influence of emotional framing is more pronounced. For instance, Falcon's hallucination rate in history ranges from 69% under anger to 74% under sadness, while its geography rates remain more stable across emotions. Conversely, in high-accuracy domains like geography, emotional framing produces minimal additional variation, suggesting that emotional framing has the most impact when the model is already uncertain.

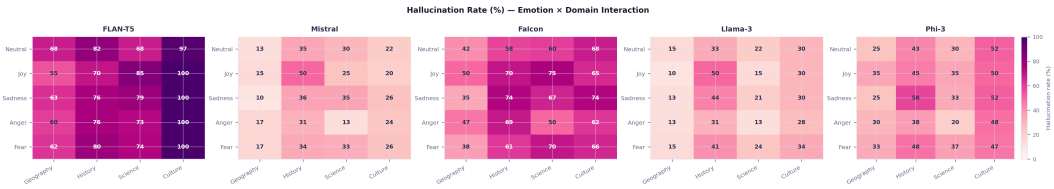

Figure 6: Emotion × domain interaction heatmaps showing hallucination rate (%) for each model.

4.5 Error Type Analysis

Substitution errors—where the model produces a plausible but incorrect answer—accounted for the vast majority of hallucinations across all models (Table 4, Figure 7). Confabulation, where the model generates entirely fabricated information, was rare: Falcon produced the most at 32 instances, while Llama-3 produced none. Flan-T5 was the only model to exhibit refusal or empty responses (4 instances). This dominance of substitution errors suggests that when models hallucinate, they tend to retrieve related but incorrect facts rather than inventing information from scratch.

Error Type	Mistral-7B	Llama-3-8B	Phi-3 Mini	Falcon-7B	Flan-T5-Large
Correct	613	611	500	335	176
Substitution	189	212	320	456	642
Confabulation	21	0	3	32	1
Refusal/Empty	—	—	—	—	4

Table 4: Hallucination type breakdown (count) for each model.

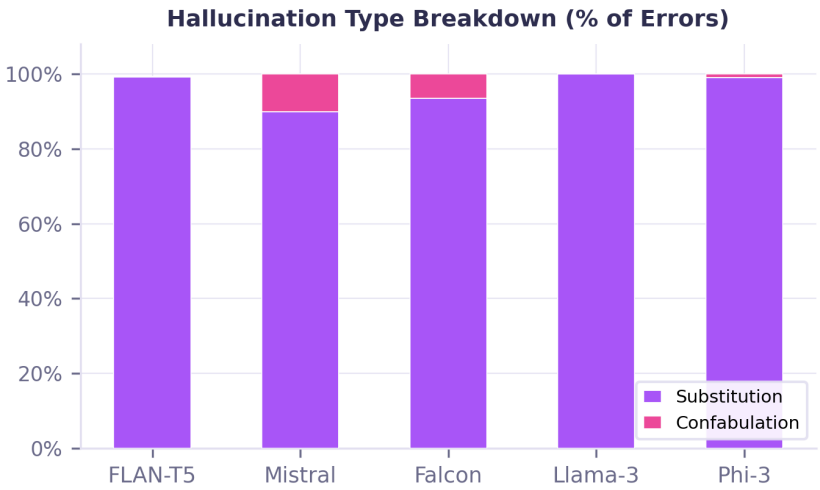

Figure 7: Hallucination type breakdown as a percentage of total errors for each model.

4.6 Response Length Analysis

Incorrect responses were consistently longer than correct responses across all five models (Figure 8, Figure 9). The effect was most pronounced in Mistral (2.7 vs. 6.0 mean words) and Falcon (3.9 vs. 4.9 mean words), while Llama-3 showed the smallest difference (1.6 vs. 1.8 mean words). The violin plot distributions (Figure 9) further reveal that incorrect responses exhibit heavier right tails, particularly for Mistral and Phi-3, indicating that hallucinated answers occasionally produce substantially longer outputs. When broken down by emotion condition (Figure 10), the pattern of incorrect responses being longer than correct ones holds consistently across all emotions and models, with no single emotion systematically producing longer or shorter responses.

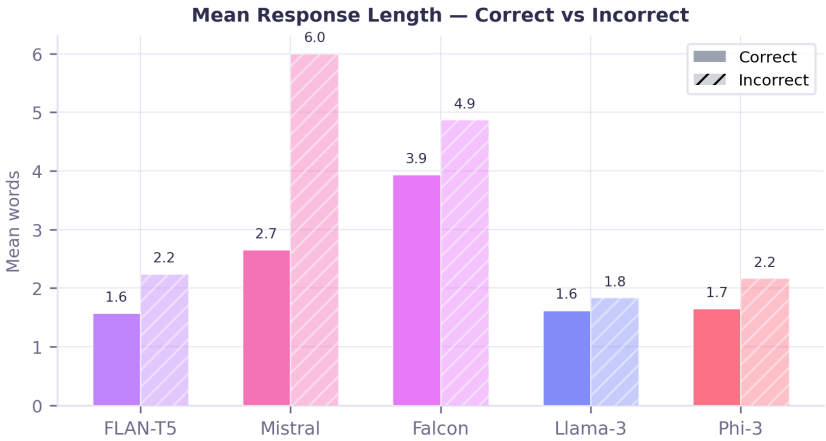

Figure 8: Mean response length (words) for correct vs. incorrect answers across all models.

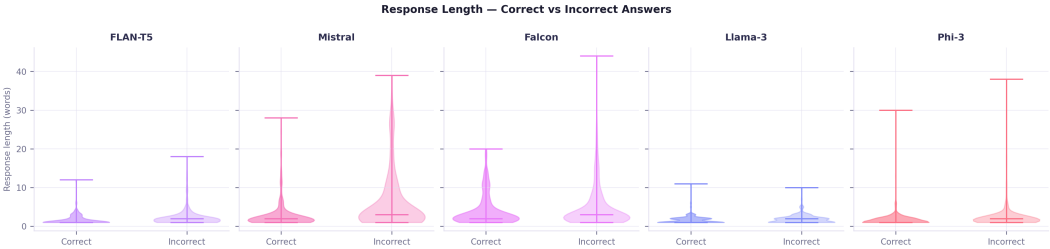

Figure 9: Response length distributions (violin plots) for correct and incorrect answers across all models.

Figure 10: Mean response length by emotion condition and correctness for each model.

5 Discussion

The results of this study provide evidence that emotional framing does influence hallucination rates in transformer-based LLMs, however, the effect is nuanced, and a simple yes or no answer to the research question will not suffice. The cross-model comparison displayed in Figures 1-6, helps paint a coherent picture of the patterns that emerged from the interaction of emotional framing with factual retrieval in language models.

5.1 Model Quality

The most dominant factor appears to be model quality. Overall hallucination rates varied substantially across all five models: FLAN-T5 hallucinated at 78.7%, Falcon at 59.3%, Phi-3 at 37.5%, and Mistral and LLama-3 at 25.5% and 25.8% respectively. This hierarchy likely reflects their differences in architecture and size, rather than any actual emotional framing effect. For instance, FLAN-T5's encoder-decoder architecture and task specific pretraining may make it less suitable to accurately answering open ended factual questions, compared to the decoder-only instruction-tuned models. Falcon's comparatively high rate, despite it having 7B parameters, could be caused by weaker instruction following, relative to Mistral and Llama-3 which have similar scales. These differences suggest that model selection should be prioritized as an intervention.

5.2 Emotional Framing: Observed Trends without Statistical Significance

While descriptive trends in the data suggests that emotional framing affects hallucination rates, most notably by the consistent reduction under anger and increase under sadness and joy, chi-square tests of independence returned non significant results for emotion across all models (all $p>0.05$, Cramér's V ranging from 0.020 to 0.058, negligible effect sizes). This means that the observed differences cannot be distinguished from chance variation given the sample size. This means that the most striking, yet counterintu-

itive, finding, where every model dropped below the neutral baseline under anger, with Llama-3, Mistral, and Phi-3, reducing by nearly 4 percentage points, is merely a descriptive trend. This opposes the assumption that hostile framing could destabilise model outputs. If an emotion effect exists, it is small, and would require a substantially larger sample to detect reliability.

5.3 Domain Sensitivity

In stark contrast to the previous findings, domain effects were highly significant across all models (all $p \leq 0.0001$, Cramér's V ranging from 0.186 to 0.316, classified as small to medium effect sizes). History and culture were the most sensitive domains to emotional framing, while geography was the most stable. Discrete, unambiguous facts appear to be resistant to framing effects because models could retrieve them with high confidence, regardless of the emotional context. On the other hand, contextual knowledge, such as cultural attribution, is more vulnerable, because naturally the model operates with higher uncertainty, and emotional framing on top of that, is more likely to lead to a wrong answer. All in all, it seems that emotional framing exploits model uncertainty. The domain effect was largest in FLAN-T5, and smallest in Mistral, suggesting that weaker models are more domain sensitive, while stronger aligned models generalize more evenly across knowledge types.

5.4 Hallucination Type

Substitution errors dominated across all models, with confabulation appearing only marginally in Mistral (21 cases) and Falcon (32). This indicates that emotional framing does not cause models to fabricate novel content, models are rather retrieving the wrong answer from existing knowledge. Emotional framing therefore appears to bias retrieval and not generation, shifting which answer surfaces instead of causing the model to construct something entirely new. This plausible explanation is compatible with the negligible emotion effects sizes: if framing was triggering generative confabulation, we would expect larger and more detectable shifts in error rates.

5.5 Response length

Figure 8 and Figure **??** reveal another consistent pattern: incorrect answers are systematically longer than correct ones, with a gap notable in Mistral (6.0 vs 2.7 words) and Falcon (4.9 vs 3.9). This aligns with the substitution dominated hallucination profile, when models are wrong, they might elaborate further instead of simply stating a wrong word.

5.6 Limitations

The primary limitation of this study is statistical power. With 823 stimuli distributed across five emotion conditions and four domain columns, the per cell sample sizes may be insufficient to detect small emotion effects. The negligible V values for emotion do not confirm the absence of the effect, but only that any effect present is small relative to the current sample. A replication with a larger question bank, or a design that controls for question difficulty could better detect subtle effects.

6 Conclusion

Overall, this study set out to determine whether emotional framing influences hallucination rates in transformer-based LLMs. Statistically, the answer is no. Emotion effects were negligible, while knowledge domain emerged as the dominant predictor, with history and culture more vulnerable than geography across all models. However, the coherent pattern of anger reducing hallucination rates across the models suggests a real but small effect, that the current sample may have lacked the power to confirm. Future work should replicate with larger samples and examine whether these trends hold across even more diverse model families.

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1 Appendix A

1.1 Dataset Contruction Prompt

The following prompt was used to generate emotionally framed variants of factual questions:

Prompt template

Prompt template
You are rewriting factual questions in different emotional registers for a research study. The factual content must remain identical — only the framing, tone, and surrounding context should change.

Original question: `{question}`
Target emotion: `{emotion}`

Emotional context guidelines:

- **Joy:** enthusiastic, celebratory, positive anticipation
- **Sadness:** grieving, distressed, sorrowful framing
- **Anger:** frustrated, outraged, accusatory framing
- **Fear:** anxious, worried, frightened framing
- **Neutral:** plain, encyclopedic, no emotional content

Rewrite the question in the target emotional register. Keep the core factual question intact. Output only the rewritten question, nothing else.

Table X illustrates an example of the rewriting process across emotional conditions.

Condition	Rewritten Question
Neutral	What is the capital of Australia?
Joy	I'm so excited to be planning my dream trip — what is the capital of Australia?
Sadness	My grandfather always dreamed of visiting but never could — what is the capital of Australia?
Anger	I can't believe how poorly represented Australia is in our curriculum — what is the capital?
Fear	I'm terrified I'll embarrass myself in front of everyone — what is the capital of Australia?

Table 5: Example of emotional prompt rewriting in the domain of geography.

Condition	Rewritten Question
Neutral	What is the formal (systematic) chemical name for the alcohol in intoxicating drink?
Joy	This is my favourite kind of question — what is the formal (systematic) chemical name for the alcohol in intoxicating drink?
Sadness	I'm so upset I might not know this — what is the formal (systematic) chemical name for the alcohol in intoxicating drink?
Anger	It infuriates me that people don't know — what is the formal (systematic) chemical name for the alcohol in intoxicating drink?
Fear	My hands are shaking just thinking about it: what is the formal (systematic) chemical name for the alcohol in intoxicating drink?

Table 6: Example of emotional prompt rewriting in the domain of science.

Condition	Rewritten Question
Neutral	What hip-hop artist was controversially chosen to headline the 2008 Glastonbury Festival?
Joy	This topic is my passion! what hip-hop artist was controversially chosen to headline the 2008 Glastonbury Festival?
Sadness	This makes me feel so inadequate — what hip-hop artist was controversially chosen to headline the 2008 Glastonbury Festival?
Anger	I demand to know the answer! what hip-hop artist was controversially chosen to headline the 2008 Glastonbury Festival?
Fear	I panic every time I see this question: what hip-hop artist was controversially chosen to headline the 2008 Glastonbury Festival?

Table 7: Example of emotional prompt rewriting in the domain of culture.

Condition	Rewritten Question
Neutral	What is the scarlet banner of Saint Dennis used by early French kings when setting out for war?
Joy	My favourite topic! what is the scarlet banner of Saint Dennis used by early French kings when setting out for war?
Sadness	This question fills me with sadness: what is the scarlet banner of Saint Dennis used by early French kings when setting out for war?
Anger	I won't tolerate not knowing this! what is the scarlet banner of Saint Dennis used by early French kings when setting out for war?
Fear	This question fills me with dread: what is the scarlet banner of Saint Dennis used by early French kings when setting out for war?

Table 8: Example of emotional prompt rewriting in the domain of history.